Fast exact overlap minimization for three-dimensional irregular strip packing of polyhedra

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Abstract

We address the three-dimensional irregular strip packing problem: a set of polyhedra, possibly non-convex, must be placed by translation in a container with a fixed rectangular base so that the height is minimized. Pieces are represented exactly as unions of convex polytopes. Overlap between two convex parts is measured by the penetration depth of their relative position in the no-fit polytope (NFP), a quantity that, along any axis direction, is concave on an interval and zero outside it. Hence the position of a piece along an axis that minimizes its weighted overlap with all other pieces can be computed exactly by evaluating interval endpoints only. We embed this move in a strategic oscillation scheme with guided local search, kicks, and a global linear-programming compaction over NFP facets. An ablation study shows the contribution of each component and that the speed of the separation is decisive. On the Merged instances of Egeblad et al. (2009), which use the same exact geometry, the method improves the best known heights by 3–9% and reaches in 10 seconds the utilization that the reference method attains in 600 seconds. It also improves the best known solutions of a classical convex instance and of the non-convex instances of Stoyan et al. (2004). Finally, a margin-based final compaction and dyadic rounding yield solutions whose feasibility is certified in exact rational arithmetic.

Keywords: Cutting, Packing, 3D irregular packing, No-fit polytope, Guided local search, Linear programming

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1. Introduction

Three-dimensional irregular packing arises in additive manufacturing (packing parts in the build volume of a printer), in the loading of non-rectangular goods, and in the design of compact assemblies. In its open-dimension variant, the *3D irregular strip packing problem*, a set of polyhedra must be placed without overlap inside a container with a fixed $W \times D$ base, and the height of the packing is to be minimized. We consider the translational version: pieces keep their orientation.

The literature on this problem is small compared with its two-dimensional counterpart, and it is split by the representation of the geometry. Exact approaches use the polyhedra directly: phi-functions and nonlinear programming (Stoyan et al., 2004, 2005; Romanova et al., 2018), the 3DNest local search of Egeblad et al. (2009), which minimizes the exact overlap volume along axis-parallel translations, constructive heuristics such as HAPE3D (Liu et al., 2015), and, recently, mixed-integer models based on no-fit polyhedra that solve small convex instances to optimality (Tollenaere et al., 2026). Approximate approaches discretize the pieces into voxels (Lamas-Fernandez et al., 2023); voxels make overlap tests trivial and scale well, but they overestimate the pieces and the quality of the solutions depends on the resolution.

In two dimensions, the most successful methods for irregular strip packing are overlap-minimization heuristics: the height (length) is decreased, the pieces are allowed to overlap, and a local search moves them until the overlap vanishes (Egeblad et al., 2007; Umetani et al., 2009; Imamichi et al., 2009; Elkeran, 2013; Gardeyn et al., 2025). The key ingredients are a cheap and informative overlap measure, moves that can be optimized exactly or quickly, and guided local search (GLS) weights to escape local minima. This paper brings these ingredients to three-dimensional polyhedra with an exact representation.

Contributions.

1. An exact and fast overlap-minimization move for polyhedra represented as unions of convex parts. Overlap is measured by the penetration depth in the NFP of each pair of convex parts; along an axis this function is concave on an interval and zero outside it, so the optimal position of a piece along an axis is attained at an interval endpoint (Proposition 1). The move extends the directional moves of Umetani et al. (2009) to 3D and non-convex pieces and is much cheaper than the exact overlap volume of Egeblad et al. (2009).

2. A global compaction by linear programming over NFP facets, in the spirit of Gomes and Oliveira (2006) in 2D, integrated in a strategic oscillation with multiplicative GLS weights (Gardeyn et al., 2025) and kicks.
3. An ablation study of the components, showing that the speed of the separation is decisive.
4. New best known solutions on the Merged instances of Egeblad et al. (2009) and on classical instances of Stoyan et al. (2004, 2005), with a time-matched comparison against the exact method 3DNest.
5. Solutions with certified feasibility: a final compaction with soft margins and dyadic rounding produce layouts that are verified in exact rational arithmetic, without tolerances.
6. An exact convex decomposition of the non-convex (prismatic) pieces of Ikonen et al. (1997) used in the Merged instances, and the identification of inconsistencies in the public data.

Section 2 defines the problem and the geometric tools. Section 3 describes the algorithm, Section 4 the certification, and Section 5 the experiments.

2. Problem and geometry

2.1. Problem

We are given a container base $W \times D$ and n pieces $P_1, \dots, P_n$ (polyhedra, with multiplicities). Each piece is a union of convex polytopes, $P_i = \bigcup_{a \in \mathcal{A}_i} A_a$, expressed in a local frame where the minimum of its bounding box is the origin; w_i, d_i, h_i denote its bounding-box dimensions. A solution is a vector of translations $t_i = (x_i, y_i, z_i)$ such that

$$0 \le x_i \le W - w_i, \quad 0 \le y_i \le D - d_i, \quad z_i \ge 0, \qquad i = 1, \dots, n, \tag{1}$$

$$\operatorname{int}(P_i + t_i) \cap \operatorname{int}(P_j + t_j) = \emptyset, \qquad i < j, \tag{2}$$

and the objective is to minimize $H = \max_i (z_i + h_i)$. A trivial lower bound is $H_0 = \max\{\max_i h_i,\ \sum_i \operatorname{vol}(P_i)/(WD)\}$.

2.2. No-fit polytopes and penetration depth

For two convex parts A and B placed at t_A and t_B, their interiors intersect if and only if the relative position $r = t_A - t_B$ lies in the interior of the no-fit polytope

$$N_{AB} = B \oplus (-A) = \operatorname{conv}\{b - a :\ a \in V(A),\ b \in V(B)\}, \tag{3}$$

where $V(\cdot)$ is the vertex set (Bennell and Oliveira, 2008). We compute N_{AB} once per pair of parts as the convex hull of the vertex differences and store it in H-representation, $N_{AB} = \{r : \ n_f \cdot r \le d_f, \ f \in F_{AB}\}$, with unit normals $\|n_f\| = 1$.

Definition 1. *The* penetration *of A and B at relative position r is* $\delta_{AB}(r) = \max\{0, \ \min_{f \in F_{AB}}(d_f - n_f \cdot r)\}$.

For r inside N_{AB}, $\delta_{AB}(r)$ is the Euclidean distance from r to the boundary of N_{AB}, i.e., the length of the shortest translation that separates A and B (the translational penetration depth). It is zero if and only if the parts do not overlap. For two pieces we use $\delta_{ij}(t_i - t_j) = \sum_{a \in \mathcal{A}_i} \sum_{b \in \mathcal{A}_j} \delta_{ab}(t_i - t_j)$, where only pairs of parts whose bounding boxes intersect are evaluated.

2.3. Exact minimization along an axis

Let all pieces but i be fixed, and let i move along axis e_k, $t_i(s) = t_i + (s - t_{ik})e_k$, with $s \in [0, U_k]$ (the container limits, using the current height H for the vertical axis). The weighted overlap of i is

$$\phi(s) = \sum_{j \neq i} w_{ij} \, \delta_{ij}\big(t_i(s) - t_j\big), \tag{4}$$

with GLS weights $w_{ij} \ge 1$.

Proposition 1. *Each term $\delta_{ab}(t_i(s) - t_j)$ is zero outside an interval (L_{ab}, U_{ab}) and concave inside it. Hence ϕ is concave on every interval between consecutive points of $\mathcal{B} = \{0, U_k\} \cup \{L_{ab}, U_{ab}\}$, and a minimizer of ϕ on $[0, U_k]$ is attained at a point of $\mathcal{B}$.*

Proof. Along the line, $d_f - n_f \cdot r(s) = c_f - n_{fk}s$ is affine. The set where all of them are positive is the open interval (L_{ab}, U_{ab}) with $L_{ab} = \max_{n_{fk} < 0} c_f / n_{fk}$ and $U_{ab} = \min_{n_{fk} > 0} c_f / n_{fk}$ (facets with $n_{fk} = 0$ only decide whether the interval is empty). Inside it, δ_{ab} is a minimum of affine functions, hence concave; outside it is zero. Between consecutive points of $\mathcal{B}$ every term is either concave or identically zero, so ϕ is concave there and attains its minimum at an endpoint. □

The move evaluates ϕ at the points of $\mathcal{B}$ only. With the interval of each term known, a term is evaluated only at the candidates inside it: sorting the candidates, each term contributes to a contiguous range found by binary search. Facets are stored as (c_f, n_{fk}) in contiguous arrays, so evaluating a term costs one fused multiply-add per facet. Ties are broken towards the smaller coordinate (gravity and walls).

Algorithm 1: Overview

```
S ← Greedy(); S ← Compact(S); H ← H(S);
h ← (1 − ρ)H; T ← positions of S clamped to height h; w ← 1;
while time remains do
    if MinimizeOverlap(T, h, w) then
        T ← Compact(T);
        if H(T) < H then S ← T; H ← H(T);
        h ← (1 − ρ)H; T ← positions of S clamped to h; w ← 1;
    else
        h ← h + ρH/4;
        if h ≥ H then
            T ← positions of S clamped to h ← (1 − ρ)H; reinsert a
              random piece of T (kick); w ← 1;
SeparateWithMargin(S); return S;
```

3. Algorithm

Algorithm 1 summarizes the method: a greedy construction, a compaction, and a strategic oscillation of the height that alternates separation attempts and kicks until the time limit.

3.1. Greedy construction

Pieces are placed one at a time on top of the current load. For a piece, $g \times g$ starting points ($g = 4$) on the base are lifted to the current height and each descends by a sequence of small linear programs in the three translation variables: for every nearby placed part one NFP facet satisfied at the current point is selected (the one that allows the longest step in the descent direction), and $z + \sigma\,\mathrm{dist}_{\mathrm{wall}}(x, y)$ is minimized within a trust region; the procedure repeats until no improvement. Among all pieces and resulting positions, the greedy chooses the one maximizing $\alpha \log(V/V_{\max}) - \beta z/H_0 - \gamma \max(0, z + h - H_{\mathrm{cur}})/H_0$, which favors large pieces placed low without increasing the height. Parameters ($\alpha = 1$, $\beta = 5.53$, $\gamma = 0.77$, $\sigma = 0.065$) were tuned with Optuna.

3.2. Overlap minimization

MinimizeOverlap performs passes over the overlapping pieces in random order; for each piece the three axes are tried in random order with

the exact move of Proposition 1. The pairwise penetrations are kept in a matrix and only the row of a moved piece is recomputed. When a pass does not decrease (4), the GLS weights are updated multiplicatively as in Gardeyn et al. (2025): the weight of every overlapping pair is multiplied by $1.2 + 0.8\,\delta_{ij}/\max\delta$, and the other weights decay by a factor 0.95 (minimum 1). The procedure stops when no overlap remains (then verified with a separating-axis test) or after 500 passes. Weights persist along the oscillation and are reset after a successful separation or a kick.

3.3. Strategic oscillation and kicks

After a successful separation the target height is decreased by $\rho = 1\%$ of the best height; after a failure it is increased by $\rho/4$ and the search continues from the overlapping layout. When the target reaches the best height, a kick restarts from the best solution at height $(1-\rho)H$ and reinserts a random piece at the lowest free position (if none exists, it swaps two pieces of different types). This scheme follows Egeblad et al. (2009) and Umetani et al. (2009).

3.4. Global LP compaction

Compact solves a sequence of linear programs over all translations. Given the current layout, for every pair of parts that can touch within a trust region (each coordinate may move at most Δ, half of the smallest piece dimension) one NFP facet f satisfied by the current relative position is selected and the constraint $n_f \cdot (t_i - t_j) \geq d_f$ is imposed. The LP

$$
\begin{aligned}
&\min\ H + \epsilon \sum_i z_i \\
&\text{s.t. } z_i + h_i \leq H, \quad \|t_i - t_i^0\|_\infty \leq \Delta, && i = 1,\dots,n, \\
&\qquad \text{container bounds on } t_i, && i = 1,\dots,n, \\
&\qquad n_f \cdot (t_i - t_j) \geq d_f, && \text{selected pairs of parts},
\end{aligned}
\tag{5}
$$

has a convex feasible region that contains the current layout. The facets are re-selected and the LP is solved again while H decreases. A solution is accepted only if it is feasible (the LP solution is checked for bounds and overlap). Compaction alternates with a local improvement in which every piece descends again with the others fixed and the pieces that define the height are tried at other positions.

4. Certified feasibility

LP solvers and floating-point NFPs leave overlaps of order 10^{-14} between touching pieces; tolerant verifiers accept them, but in exact arithmetic the pieces overlap. Note that safe bounds for linear programs (Neumaier and Shcherbina, 2004) certify the optimal value of the LP, not the feasibility of the point returned by the solver, so we certify the final layout instead.

SeparateWithMargin solves a last compaction LP in which each pair constraint becomes $n_f \cdot (t_i - t_j) \geq d_f + m_k$ and each container bound $x_i + u_i \leq W - w_i$ (and likewise in y), with slacks $m_k, u_i \in [0, \mu]$ ($\mu = 10^{-7}$, well above the LP tolerances) that are maximized while the height is kept. The slacks must be soft: rows of pieces that exactly fill the width of the container leave no room for a margin. In those rows the contact positions are typically sums of piece dimensions, which are representable; we then round every coordinate to the nearest multiple of 2^{-30}, an exact operation in binary floating point whose error is far smaller than μ.

The certificate is computed with rational arithmetic: vertices and translations are binary floating-point numbers, hence exact rationals. For containment all vertices are checked against the container; for every pair of convex parts with intersecting bounding boxes a separating axis is searched among the face normals of both parts and the cross products of their edges (computed exactly), accepting it if $\max_A n \cdot a \leq \min_B n \cdot b$. In 22 test runs (11 instances, 2 seeds, 20 s) all solutions were certified; the height increases by 10^{-7}–10^{-6}.

5. Computational experiments

5.1. Setting

The method is implemented in C++ on top of the Metasolver framework; LPs are solved with SoPlex (Wunderling, 1996) through the ibex library (Chabert et al., 2024). Experiments ran on an Intel Xeon at 2.8 GHz, one thread per run, four runs in parallel. Each configuration was run with 10 seeds and a time limit of 120 s per instance.

Instances. We use the 3D instances of the ESICUP repository (EURO Special Interest Group on Cutting and Packing, 2025):

- *Stoyan 2005*, Examples 1–3 (7, 12 and 25 convex pieces) (Stoyan et al., 2005). In the public data, one piece (polytope7) has faces that use

only 9 of its 10 vertices; we use its convex hull (volume 192.5), which matches the volume reported by Egeblad et al. (2009).

- *Stoyan 2004*, Examples 1–5 (20–50 non-convex pieces) (Stoyan et al., 2004), each defined as a union of convex polytopes, which we use directly as the decomposition.

- *Merged1–5* (Egeblad et al., 2009): i copies of 15 pieces, the 7 convex pieces of Stoyan 2005 and the 8 pieces of Ikonen et al. (1997) scaled by 4, in a container whose base is such that a layout with 50% utilization is a cube. Six Ikonen pieces are non-convex prisms; we decompose them exactly by triangulating the base polygon, merging triangles into convex polygons (Hertel and Mehlhorn, 1983), and extruding (a frame is split into four boxes; two pieces have polygons touching at a vertex). The volumes of the parts match those of the meshes.

References. 3DNest (Egeblad et al., 2009) uses the same exact geometry; its average and maximum utilization over 10 runs of 600 s (one core of a 2.16 GHz Core Duo) are converted to heights. Lamas-Fernandez et al. (2023) report best and average heights of a voxel VNS; their voxelization overestimates the pieces (by 15% in volume for Merged1, as they note), so their heights are not directly comparable. For Stoyan 2004 the only exact reference is Stoyan et al. (2004).

5.2. Component analysis

Table 1 evaluates the components on the seven hardest instances (Stoyan 2004 Examples 1–5 and Stoyan 2005 Examples 2–3). Each row changes one element of the method. The effect of the speed of the separation is estimated by stopping the same runs at 24 s, which corresponds to moves five times slower.

5.3. Comparison with the literature

Table 2 compares the method with the literature. On the instances that share the exact geometry of 3DNest (Stoyan 2005 Examples 2–3 and Merged1–5), the method obtains lower heights in six of seven cases; the exception is Stoyan 2005 Example 2 (19.13 versus 19.01). The improvement grows with the size of the instance: on Merged3–5 our best heights are 8–9% below the best known values, obtained by the voxel VNS with the finest

Table 1: Mean gap of the height with respect to the best published value (Stoyan 2004 Examples 2–5 and Stoyan 2005 Examples 2–3) and mean height on Stoyan 2004 Example 1 (no published reference); 120 s, mean of 10 seeds. Negative is better.

Configuration	Stoyan 2004	Stoyan 2005	S2004 Ex1
Proposed method	TBD	TBD	TBD
Without LP compaction	TBD	TBD	TBD
Additive GLS weights ($w_{ij} \leftarrow w_{ij}+1$)	TBD	TBD	TBD
Kick: swap two pieces instead of reinserting one	TBD	TBD	TBD
Separation five times slower (runs stopped at 24 s)	TBD	TBD	TBD

resolution and four hours. On Stoyan 2004, the method improves the exact reference by a wide margin, and on Example 2 it reaches the optimum.

On Stoyan 2004 Example 2 our height, 29, equals the lower bound H_0 given by the tallest piece, so it is optimal for the orientation of the pieces in the public data. The height of 28.47 reported by Lamas-Fernandez et al. (2023) is below the height of that piece; since their orientations are also fixed, their meshes must differ from the public data, most likely in the orientation of some pieces (the piece has dimensions $4 \times 26 \times 29$ and could lie with its longest side horizontal). The comparison with the voxel results on Stoyan 2004 should therefore be taken with caution, whereas the instances shared with 3DNest were checked against the volumes, bounding boxes and container dimensions reported by Egeblad et al. (2009). Stoyan 2004 Example 1 is not included: the instance used by Lamas-Fernandez et al. (2023) under that name is a different set of pieces (Liu et al., 2015). For Stoyan 2005 Example 1, the utilization reported by Egeblad et al. (2009) converts to a height below the proven optimum with our data, so the instance seems to differ and we only compare with the optimum. Figure 1 shows two of the best solutions.

5.4. Time-matched comparison

Hardware differs by about fifteen years. Figure 2 compares the mean utilization over time with that of 3DNest at 10, 60, 300 and 600 seconds. Our first solution is available after about 6 seconds (the greedy construction and compaction), and at 10 seconds our mean utilization already exceeds the mean utilization of 3DNest at 600 seconds on all five instances. Even

Table 2: Heights (lower is better). Exact: 3DNest (Egeblad et al., 2009) (best and mean of 10 runs of 600 s), optimum of Tollenaere et al. (2026) for Stoyan 2005 Example 1, and Stoyan et al. (2004) (St04). Voxels: VNS of Lamas-Fernandez et al. (2023) (best: standard resolution and 10 runs of 1 h for Stoyan, extra-fine resolution and 4 h for Merged; mean: standard resolution). Ours: mean (best) of 10 seeds, 120 s. *Optimal (equal to the lower bound H_0). †Below the height of the tallest piece (29) in the public data; the instance probably differs in the orientation of the pieces, see the text.

		Exact		Voxels		
Instance	n	best	mean	best	mean	Ours
Stoyan 2005 Ex1	7	17.46 (opt.)	–	–	–	17.48 (17.46)
Stoyan 2005 Ex2	12	**19.01**	19.83	21.23	21.70	19.13 (19.13)
Stoyan 2005 Ex3	25	28.83	29.82	29.76	31.53	**27.52 (27.00)**
Stoyan 2004 Ex2	20	32 (St04)	–	28.47†	29.40	**29.00 (29.00)***
Stoyan 2004 Ex3	30	49 (St04)	–	41.30	42.65	**40.15 (40.00)**
Stoyan 2004 Ex4	40	63.21 (St04)	–	54.83	59.52	**52.69 (50.91)**
Stoyan 2004 Ex5	50	78.7 (St04)	–	72.33	76.81	**67.21 (66.54)**
Merged1	15	17.00	17.45	17.13	18.67	**16.60 (16.55)**
Merged2	30	21.98	22.02	22.06	22.64	**21.00 (21.00)**
Merged3	45	24.04	24.97	24.32	26.26	**22.07 (22.00)**
Merged4	60	27.81	28.35	26.47	29.44	**24.47 (24.01)**
Merged5	75	30.13	31.31	29.24	31.12	**27.04 (27.00)**

assuming that our machine is five times faster per core, the advantage holds at equal effective time.

5.5. Discussion

The results support a conclusion that agrees with recent 2D work (Gardeyn et al., 2025): the quality of an overlap-minimization heuristic is driven by how many separation attempts fit in the time budget, and therefore by the cost of evaluating and optimizing the overlap. The NFP-based penetration depth gives an exact, cheap and informative measure for unions of convex parts. The global LP compaction complements the axis moves, which cannot produce coordinated or diagonal displacements.

Limitations. Rotations are not considered. The method requires a convex decomposition of each piece; for the instances used here it is exact and small, but general meshes may need many parts, and approximate decompositions

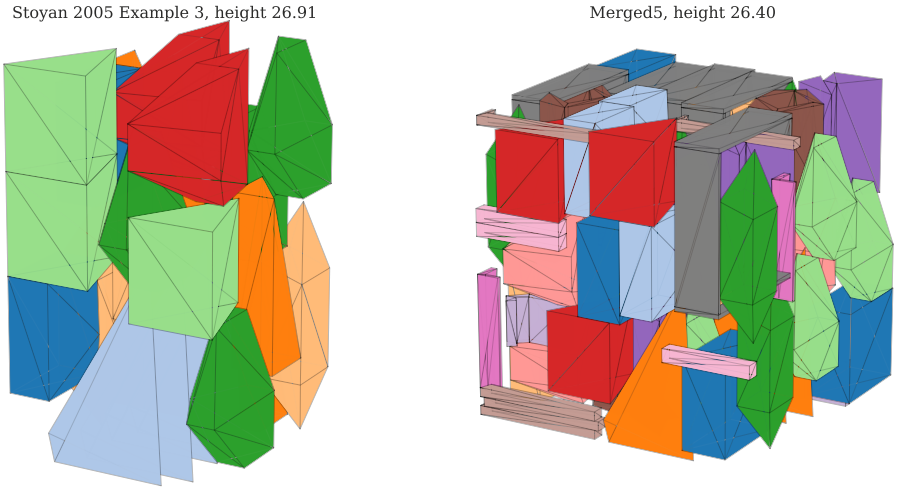


Figure 1: Best solutions found for Stoyan 2005 Example 3 and Merged5.

lose the exactness of the geometry. [TODO: instancias adicionales (Tollenaere et al., 2026) y mas grandes]

6. Conclusions

We presented an overlap-minimization heuristic for 3D irregular strip packing that works with the exact geometry of polyhedra given as unions of convex parts. Its key element is an exact axis move based on penetration depths in no-fit polytopes, combined with multiplicative guided local search, strategic oscillation and LP compaction. A simple core of these components outperforms the previous methods, with exact and voxel geometry, on the classical benchmarks, and its solutions can be certified in exact arithmetic. Future work includes rotations, larger instances from additive manufacturing, and certified lower bounds.

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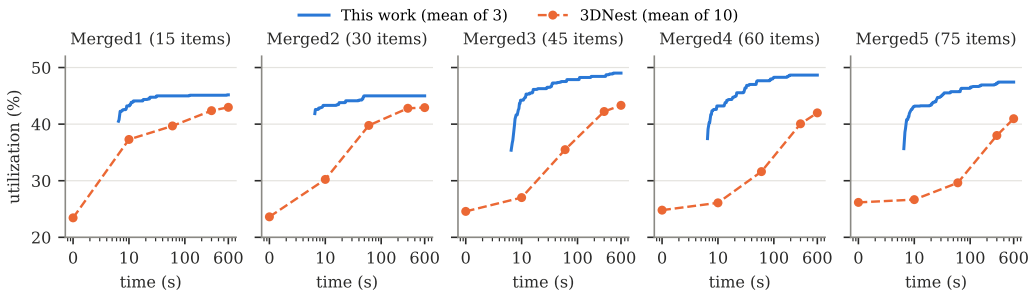


Figure 2: Mean utilization over time on the Merged instances (10 seeds), compared with 3DNest (Egeblad et al., 2009, Table 3) (10 runs).

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